GRILSS: Global Reservoir Inventory of Lost Storage by Sedimentation

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Overview

The **Global Reservoir Inventory of Lost Storage by Sedimentation (GRILSS)** dataset provides open-source data on sedimentation and capacity loss for 1,013 reservoirs in 75 major river basins across 54 countries. This dataset serves as a critical resource for studying reservoir sedimentation, which affects the storage capacity of artificial reservoirs, with implications for water management, environmental sustainability, and infrastructure planning.

The dataset consists of two main components: sedimentation data and spatial vector data, allowing for detailed geospatial analysis of reservoirs, dams, and catchments worldwide. The data is intended to support sediment management strategies, research, and future contributions from the community.

File Structure

`GRILSS_v1.2.zip` contains the following directories and files:

1. **Sedimentation_data**
    - `GRILSS_data_v1.2.xlsx` : Contains sedimentation data, with detailed fields described in the table below.
2. **Vector_data**
    - **geojson** folder:
        - `GRILSS_reservoirs_v1.2.geojson` : Reservoir boundary polygons.
        - `GRILSS_dams_v1.2.geojson` : Dam point locations.
        - `GRILSS_catchments_v1.2.geojson` : Catchment polygons.
        - `GRILSS_dam_sp_pts_v1.2.geojson` : Snap point locations corresponding to dam locations to create catchment polygons.
    - **shapefile** folder:
        - `/reservoirs_shapefile/GRILSS_reservoirs_v1.2.shp` : Reservoir boundary polygons (Shapefile format).

- `/dams_shapefile/GRILSS_dams_v1.2.shp`: Dam point locations (Shapefile format).
- `/catchments_shapefile/GRILSS_catchments_v1.2.shp`: Catchment polygons (Shapefile format).
- `/catchments_shapefile/GRILSS_dam_sp_pts_v1.2.shp`: Snap point locations corresponding to dam locations to create catchment polygons. (Shapefile format).

GRILSS_data_v1.2.xlsx Field Descriptions

Field Name	Description	Units	Data Type
GRILSS OID	Unique identifier for each sedimentation record	-	Integer
GRILSS RID	Unique identifier for each reservoir	-	Integer
Reservoir	Name of the reservoir	-	String
Country	Country where the reservoir is located	-	String
HYBAS_ID	HydroSHEDS ID for the major basin containing the reservoir	-	Integer
Major River Basin	Name of the major river basin (unique to each HYBAS_ID)	-	String
Continent	Continent where the reservoir is located	-	String
Capacity Loss Rate (%/year)	Annual rate of capacity loss	%/year	Float
Sedimentation Rate (MCM/year)	Annual sedimentation rate	Million cubic meters/year	Float
Capacity Loss (%)	Percentage of original capacity lost	%	Float
Sedimentation Amount (MT)	Total sediment deposited	Million tons	Float
Sedimentation Amount (MCM)	Volume of sediment deposited	Million cubic meters	Float
Sediment Bulk Density (ton/m^3)	Bulk density of sediment	Tons/cubic meter	Float
Observed Duration (years)	Duration over which sedimentation was measured	Years	Integer
Observation	Year when sedimentation observation	-	Integer

End Year	ended		
Observation Start Year	Year when sedimentation observation began	-	Integer
Built Year	Year of reservoir construction	-	Integer
Original Built Capacity (MCM)	Original capacity at time of construction	Million cubic meters	Float
Catchment Area (km^2)	Catchment area of the reservoir	Square kilometers	Float
Height (m)	Height of the reservoir's impoundment	Meters	Float
Latitude	Latitude coordinate of the dam	Degrees	Float
Longitude	Longitude coordinate of the dam	Degrees	Float
Survey Type	Method used to estimate sedimentation volume (BS-Bathymetric Survey, RS-Remote Sensing, NM-Numerical Modeling)	-	String
Type of Storage	Type of reservoir storage for sedimentation estimate (Gross or Live)	-	String
Comments	Additional information	-	String
Original Source URL	URL of the original data source	-	String
Fields Corrected	Names of fields manually corrected	-	String
Fields Computed	Names of fields computed for this record	-	String
GDAT Dam Name	Name of the dam in the GDAT dataset	-	String
GDAT Feature_ID	Feature ID in the GDAT dataset	-	Integer
GRAND_ID	Reservoir ID in the GRanD database	-	Integer
GRAND Wrong Location	Flag indicating incorrect dam coordinates in GRanD	-	Boolean (0/1)
GDAT Wrong Location	Flag indicating incorrect dam coordinates in GDAT	-	Boolean (0/1)
Dam Removed or Dried	Flag indicating if the reservoir is removed or dried	-	Boolean (0/1)
Creek Dam	Flag indicating if the dam is classified as a creek dam	-	Boolean (0/1)

Vector Data Field Descriptions

1. **Reservoir Vector Data (GRILSS_reservoirs_v1.2.geojson/.shp)**
    - GRILSS RID: Unique reservoir ID.
    - Reservoir: Reservoir name.
    - Country: Country of the reservoir.
    - Cap (MCM): Original built capacity in Million Cubic Meters.
    - Height (m): Height of the reservoir impoundment in meters.
    - Geometry: Polygon representing reservoir boundaries.
2. **Dam Vector Data (GRILSS_dams_v1.2.geojson/.shp)**
    - GRILSS RID: Unique reservoir ID.
    - Reservoir: Reservoir name.
    - Country: Country of the dam.
    - Latitude: Latitude of the dam.
    - Longitude: Longitude of the dam.
    - Geometry: Point representing the dam location.
3. **Catchment Vector Data (GRILSS_catchments_v1.2.geojson/.shp)**
    - GRILSS RID: Unique reservoir ID.
    - Reservoir: Reservoir name.
    - Country: Country of the catchment.
    - CA_AreaKm2: Computed catchment area in square kilometers (can be different from Catchment Area (Km^2) in GRILSS_data.xlsx).
    - Geometry: Polygon representing catchment boundaries.
4. **Dam Snap Point Vector Data (GRILSS_dam_sp_pts_v1.2.geojson/.shp)**
    - GRILSS RID: Unique reservoir ID.
    - Reservoir: Reservoir name.
    - Country: Country of the dam.
    - Latitude: Latitude of the dam.
    - Longitude: Longitude of the dam.
    - Geometry: Point representing the snap point for the dam to create catchment polygons.

Usage Instructions

1. **Download and Extract**: Download the `GRILSS_v1.2.zip` file and extract its contents.
2. **Sedimentation Data**: Use `GRILSS_data_v1.2.xlsx` for detailed sedimentation records. Open it in Excel or any compatible software.
3. **Vector Data**: Import the GeoJSON or Shapefile data into GIS software (such as QGIS or ArcGIS) for geospatial analysis of reservoir boundaries, dams, and catchment areas.

Applications

- **Reservoir Management**: Analyze trends in sedimentation and capacity loss for informed management decisions.

- **Global and Regional Studies**: Examine sedimentation patterns across different regions, river basins, and countries.
- **Environmental Impact Research**: Investigate how land use changes and climate variations influence sedimentation in reservoirs.

Version Updates

- v1.1 (2025-02-10): Initial release.
- v1.2 (2025-09-26): Corrected catchments polygons for some reservoirs.

License

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Citation

If you use this dataset in your work, please cite it as follows:

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